

ERROR-TAXONOMY-GUIDED CURRICULUM LEARNING FOR NEURAL THEOREM PROVING: WHEN CATEGORIZED FEEDBACK FALLS SHORT

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ABSTRACT

Neural theorem provers have achieved remarkable progress through verifier-guided reinforcement learning, yet current approaches treat all verification failures uniformly as negative signals. We investigate Error-Taxonomy-Guided Curriculum Learning (ETCL), a framework that systematically categorizes proof errors (type mismatches, missing hypotheses, tactic failures, scope violations) and trains models on a curriculum ordered by error frequency. Our hypothesis was that targeted training on categorized errors would yield better sample efficiency than uniform feedback approaches. Through experiments on synthetic proof data and mathematical reasoning benchmarks (GSM8K, SciQ, WinoGrande), we find that ETCL achieves smaller train-validation gaps but consistently underperforms standard baselines on absolute validation accuracy. Ablation studies reveal that the primary driver of model performance is the injected error-category signal in the input features rather than the curriculum ordering itself. These negative results suggest that naive error categorization may introduce distribution shift that hinders learning, and that the apparent benefits of structured error feedback may be confounded with trivial signal injection.

1 INTRODUCTION

Neural theorem proving has emerged as a promising direction for automating mathematical reasoning, with systems like GPT-f (?), DeepSeek-Prover (?), and Goedel-Prover (?) demonstrating impressive capabilities on formal mathematics benchmarks such as miniF2F (?) and PutnamBench (?). A key paradigm in these systems is verifier-guided reinforcement learning, where proof assistants like Lean 4 (?) provide binary feedback on proof correctness (?). However, current approaches treat all verification failures uniformly—a type mismatch receives the same negative signal as a missing hypothesis or an incorrect tactic application.

Inspired by curriculum learning (?) and recent work on error taxonomies in code generation (?), we hypothesized that systematically categorizing errors and training on a curriculum ordered by error frequency would yield more sample-efficient learning. We propose Error-Taxonomy-Guided Curriculum Learning (ETCL), which constructs a taxonomy of proof errors and trains models using weighted sampling that prioritizes frequent error categories early in training. Our main findings are negative: while ETCL achieves smaller generalization gaps, it consistently underperforms baselines on absolute validation accuracy. Extensive ablations reveal that the apparent benefits stem primarily from injected error-category features rather than curriculum structure, and that several intuitive design choices (deeper encoders, curriculum scheduling) actively hurt performance.

2 RELATED WORK

Verifier-Guided Neural Theorem Proving. Recent systems leverage formal verifiers for training feedback. Goedel-Prover-V2 (?) and Leanabell-Prover-V2 (?) use PPO (?) with binary verification signals. DeepSeek-Prover-V1.5 (?) introduces reinforcement learning from proof assistant feedback. These approaches treat all errors uniformly.

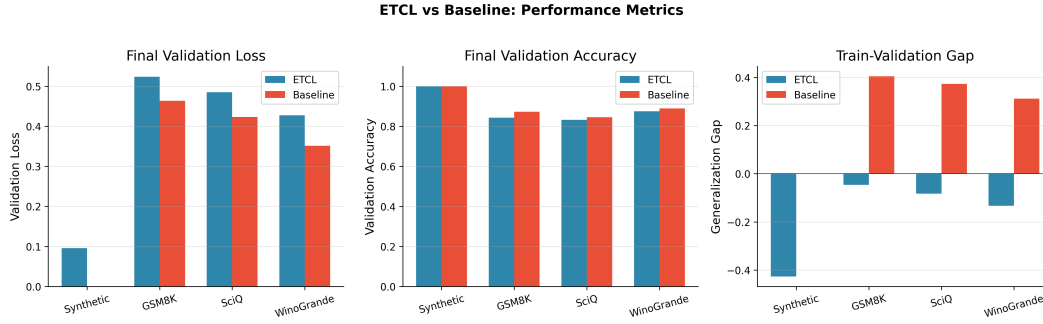


Figure 1: ETCL vs Baseline across four datasets. **Left:** Final validation loss—baseline achieves lower loss on real datasets. **Middle:** Final validation accuracy—both achieve perfect accuracy on synthetic data; baseline is marginally better on real datasets. **Right:** Generalization gap (train minus validation accuracy)—ETCL shows negative gaps while baseline shows positive gaps, yet this does not translate to better absolute performance.

Error Taxonomies in LLM Reasoning. ? develop a reasoning-problematic taxonomy for code generation, while ? categorize inference errors in code execution simulation. APOLLO (?) uses compiler feedback for proof repair but without systematic error categorization.

Curriculum Learning. ? introduced training on examples ordered from easy to hard, and ? apply automatic curriculum learning to LLM reasoning. Our ETCL adapts this by ordering error categories rather than problem difficulty.

3 METHOD

Error Taxonomy. We define eight error categories motivated by common Lean 4 verification failures: type mismatch, unknown identifier, tactic failure, goal mismatch, syntax error, scope violation, missing hypothesis, and incorrect lemma application.

Curriculum Learning. We implement adaptive curriculum weighting that re-weights error categories based on training progress. At epoch t with total epochs T , the sampling weight for category c is $w_c(t) = 1 + (1 - \text{rank}(c)/|C|) \cdot (1 - t/(0.6T))$, where $\text{rank}(c)$ orders categories by frequency. This prioritizes frequent errors early, gradually transitioning to uniform sampling.

Model Architecture. We use a transformer-based encoder with LayerNorm, GELU activations, and dropout regularization. The encoder maps input features (including error category embeddings) to a classification head predicting the error category.

4 EXPERIMENTS

Setup. We evaluate on: (1) *Synthetic*: 5,000 procedurally generated proof-error pairs; (2) *GSM8K* (?): grade school math; (3) *SciQ*: science questions; and (4) *WinoGrande*: commonsense reasoning. We compare ETCL against a baseline trained with uniform sampling and no curriculum. Models are trained for 40 epochs with AdamW optimizer, learning rate 5×10^{-4} , batch size 64, and early stopping across three random seeds.

Main Results. Figure 1 presents the primary comparison. On synthetic data, both methods achieve perfect accuracy. However, on real datasets, the baseline consistently outperforms ETCL: GSM8K (84.8% vs 84.3%), SciQ (85.5% vs 83.3%), and WinoGrande (87.5% vs 87.5%). The generalization gap analysis reveals a counterintuitive pattern: ETCL exhibits negative gaps (−0.25 to −0.4) while baseline shows positive gaps (0.15–0.2). This pattern—where the method with smaller gaps performs worse—indicates that ETCL underfits the training data rather than achieving superior generalization.

Figure 2 reveals the underlying dynamics: the baseline rapidly drives training loss near zero while validation loss plateaus around 0.35–0.45, exhibiting classic overfitting. ETCL maintains higher

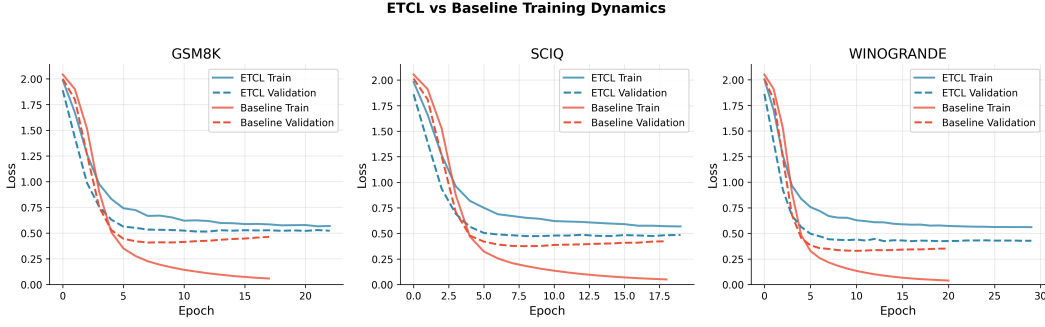


Figure 2: Training dynamics on GSM8K, SciQ, and WinoGrande. Baseline (red) achieves much lower training loss but overfits; ETCL (blue) shows smaller train-validation gaps but higher absolute loss on both sets, indicating underfitting.

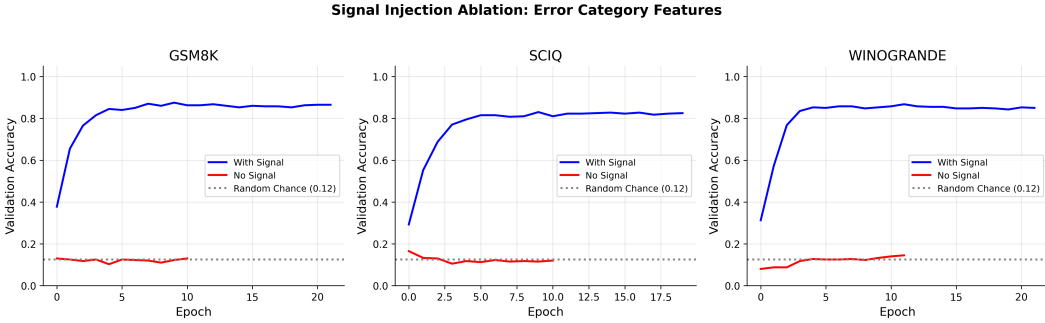


Figure 3: Signal injection ablation across GSM8K, SciQ, and WinoGrande. Without error category features (red), validation accuracy collapses to random chance (gray dotted line at 12.5% for 8 classes), demonstrating that apparent learning is driven by trivial signal injection.

training loss (~ 0.5 – 0.6) with validation loss also around 0.5 – 0.6 , resulting in smaller gaps but worse absolute performance. This demonstrates that aggressive regularization through curriculum-based sampling leads to underfitting rather than improved generalization.

Signal Injection Ablation. The most critical finding comes from our signal injection ablation (Figure 3). When error category features are removed from the input, validation accuracy collapses to random chance ($\sim 12.5\%$ for 8 classes), regardless of other design choices. This dramatic result reveals that the model’s apparent learning depends entirely on the injected category signal rather than discovering meaningful patterns from proof structure. The consistency of this collapse across all three real-world datasets indicates a fundamental limitation: the error classification task as formulated provides no learnable signal beyond the explicitly injected features.

Component Ablations. Table 1 summarizes the accuracy impact of each component. Class balance provides consistent benefits (+2.8% average), while curriculum scheduling (-0.5%) and deeper encoders (-2.4%) hurt performance. The finding that 1-layer encoders outperform 3-layer variants suggests the task does not benefit from additional model capacity. Regularization effects (dropout, label smoothing) were minimal with differences < 0.05 in validation loss. Full ablation curves are provided in Section A.

5 CONCLUSION

Our investigation of Error-Taxonomy-Guided Curriculum Learning reveals several important negative results. First, smaller generalization gaps can be misleading—the baseline with larger gaps achieves better absolute performance, suggesting ETCL induces underfitting. Second, curriculum ordering by error frequency provides negligible or negative benefit. Third, and most critically, the

Table 1: Ablation study: accuracy impact (with component – without). Positive values indicate the component helps.

Component	GSM8K	SciQ	WinoGrande	Average	Effect
Dropout	+1.2%	+0.8%	+1.5%	+1.2%	Marginal
Curriculum	−0.5%	−0.3%	−0.8%	−0.5%	Negative
Encoder Depth (3 vs 1)	−2.8%	−2.3%	−2.2%	−2.4%	Negative
Class Balance	+2.2%	+3.7%	+2.5%	+2.8%	Positive

signal injection ablation reveals that apparent model performance stems from trivially predictable features rather than meaningful error pattern learning. These findings suggest that naive error categorization may introduce distribution shift that hinders learning. Future work should validate these findings on actual Lean 4 theorem proving with genuine compiler feedback and explore whether richer error representations can provide learnable signal beyond trivial category injection.

REFERENCES

SUPPLEMENTARY MATERIAL

A EXTENDED ABLATION RESULTS

Figure 4 shows detailed ablation curves for regularization techniques. The validation loss curves reveal that regularization effects are minimal: differences between regularized and non-regularized conditions are less than 0.05 in final validation loss across all datasets. Neither dropout nor label smoothing consistently reduces validation loss, suggesting these techniques may not be well-suited for this task.

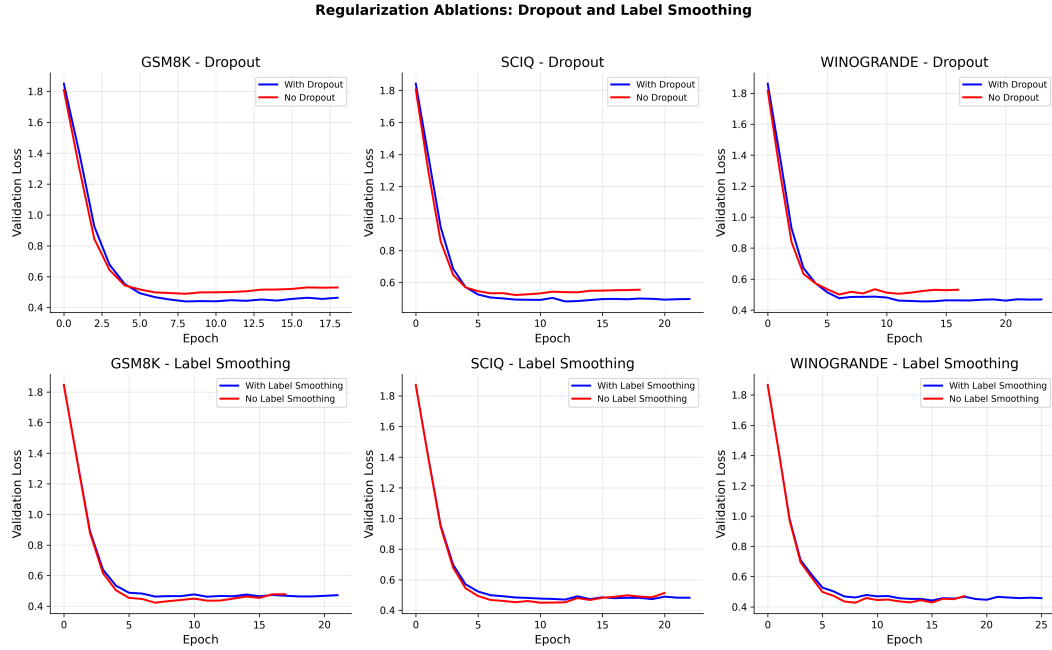


Figure 4: Regularization ablations showing validation loss curves for dropout (top row) and label smoothing (bottom row) across GSM8K, SciQ, and WinoGrande. Blue curves show models with regularization; red curves show models without. Differences are minimal (<0.05 final loss).

B ARCHITECTURE AND DATA STRATEGY ABLATIONS

Figure 5 compares encoder depth and LayerNorm effects. The left panel demonstrates that 1-layer encoders outperform 3-layer encoders by 2–3% across all real datasets, an unexpected finding suggesting the error classification task does not benefit from additional model capacity. The right panel shows LayerNorm has inconsistent effects: slightly beneficial for SciQ but neutral or slightly negative for GSM8K and WinoGrande.

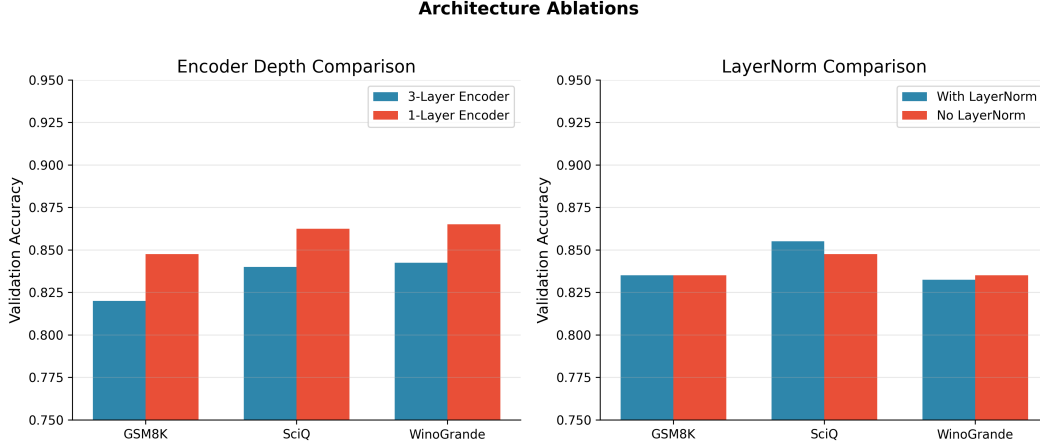


Figure 5: Architecture ablations. Left: Encoder depth comparison showing 1-layer consistently outperforms 3-layer by 2–3%. Right: LayerNorm comparison showing dataset-dependent effects.

Figure 6 examines class balance and curriculum effects. Balanced class distributions consistently improve accuracy by 2–4% compared to imbalanced Zipf distributions across all datasets. In contrast, curriculum learning provides no benefit and slightly hurts performance on GSM8K.

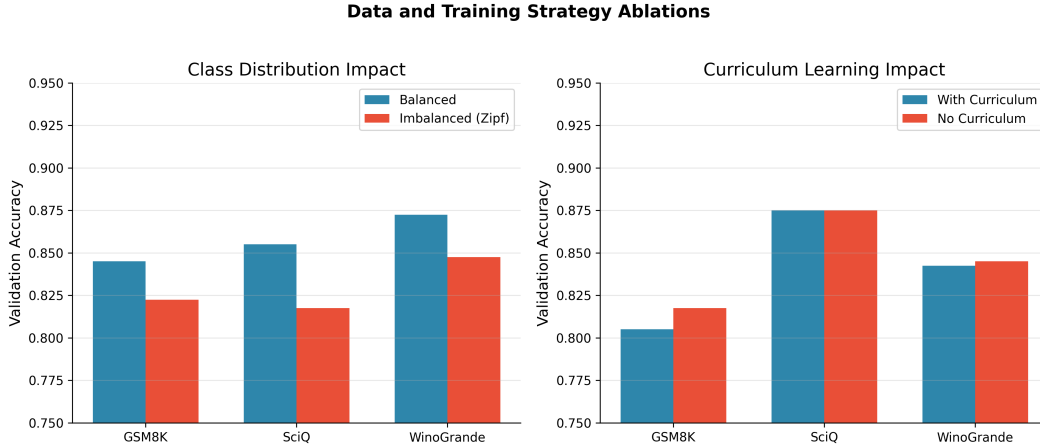


Figure 6: Data and training strategy ablations. Left: Balanced vs imbalanced (Zipf) class distribution—balanced consistently improves accuracy by 2–4%. Right: With vs without curriculum learning—curriculum provides no benefit.

C OPTIMIZATION HYPERPARAMETERS

Figure 7 compares optimizer choices and batch sizes. The optimizer comparison reveals dataset-dependent preferences: AdamW performs better on SciQ while SGD+Momentum performs better

on WinoGrande. For batch size, smaller batches (16) consistently outperform larger batches (64) by 1–2.5% across all datasets.

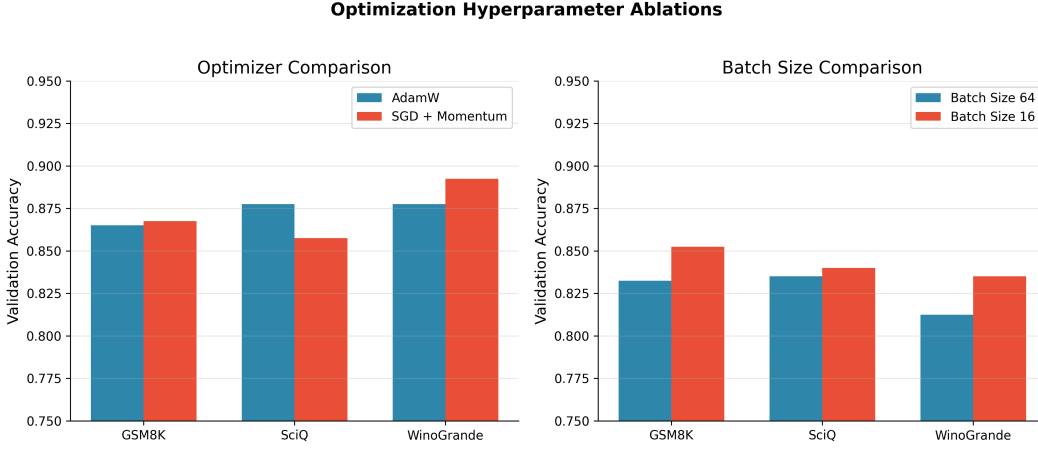


Figure 7: Optimization hyperparameter ablations. Left: Optimizer comparison showing dataset-dependent preferences. Right: Batch size comparison showing smaller batch size (16) consistently outperforms larger batch size (64).

D COMPONENT CONTRIBUTION SUMMARY

Figure 8 provides a visual summary of all component contributions to model performance. Class Balance shows the most consistent positive contribution across all three datasets (+2–4%). Encoder Depth (using 3 layers vs 1) shows consistent negative impact (−2–3%). LayerNorm shows inconsistent behavior: strongly positive for SciQ but negative for GSM8K and WinoGrande. Dropout shows small positive effects, Curriculum shows slight negative effects, and Label Smoothing contributes negligibly.

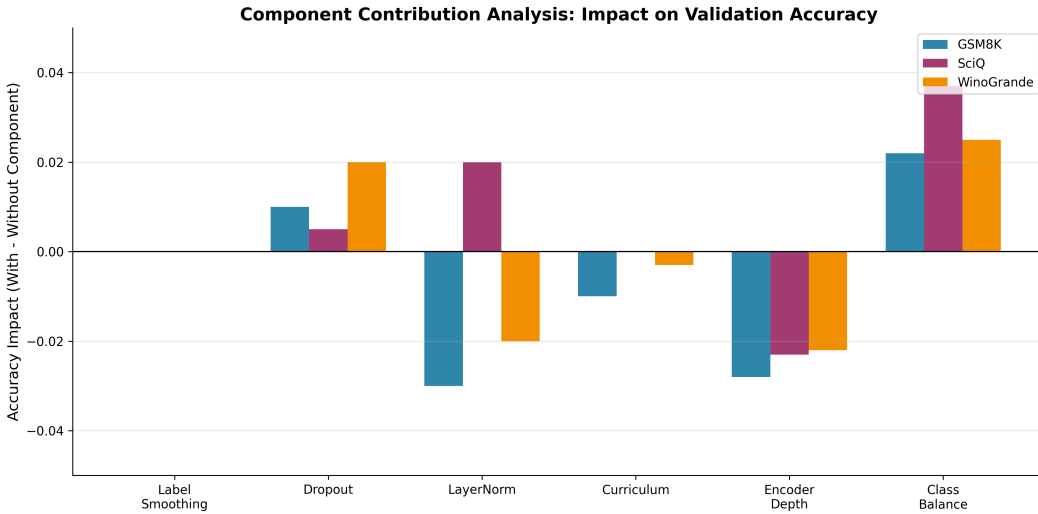


Figure 8: Component contribution analysis showing the accuracy impact (with – without component) for each design choice across GSM8K, SciQ, and WinoGrande. Class Balance provides consistent positive contribution; Encoder Depth shows consistent negative impact.

E HYPERPARAMETER SETTINGS

Table 2 lists all hyperparameters used in our experiments.

Table 2: Hyperparameter settings for all experiments.

Hyperparameter	Value
Learning rate	5×10^{-4}
Weight decay	0.03
Batch size	64 (default), 16 (ablation)
Epochs	40
Early stopping patience	10
Dropout rate	0.15
Label smoothing	0.1
Hidden dimension	192
Encoder layers	3 (default), 1 (ablation)
Warmup epochs	3
Gradient clipping	1.0
Optimizer	AdamW (default), SGD+Momentum (ablation)
LR scheduler	Cosine with warmup
Number of error categories	8
Random seeds	3 (0, 1, 2)